

Fourth Degree Function Report

Polynomial

$$f(q) = aq^4 + bq^3 + cq^2 + dq + e$$

Naive

$$g_{\text{naive}}(X) = X^4a + X^3b + X^2c + Xd + e$$

$$\mathbb{E}[g_{\text{naive}}(X)] = 24a\frac{\Delta^4}{\epsilon} + aq^4 + bq^3 + 2\frac{\Delta^2}{\epsilon}(6aq^2 + 3bq + c) + cq^2 + dq + e$$

$$\begin{aligned}\text{Var}(g_{\text{naive}}(X)) = & 2\frac{\Delta^2}{\epsilon} \left(19872a^2\frac{\Delta^6}{\epsilon} + 16a^2q^6 + 24abq^5 + 16acq^4 + 8adq^3 + 9b^2q^4 \right. \\ & + 12bcq^3 + 6bdq^2 + 24\frac{\Delta^4}{\epsilon}(408a^2q^2 + 204abq + 28ac + 15b^2) \\ & + 2\frac{\Delta^2}{\epsilon}(372a^2q^4 + 372abq^3 + 156acq^2 + 48adq + 81b^2q^2 \\ & \left. + 54bcq + 12bd + 5c^2) + 4c^2q^2 + 4cdq + d^2 \right)\end{aligned}$$

$$\begin{aligned}\text{MSE}(g_{\text{naive}}) = & 2\frac{\Delta^2}{\epsilon} \left(20160a^2\frac{\Delta^6}{\epsilon} + 16a^2q^6 + 24abq^5 + 16acq^4 + 8adq^3 + 9b^2q^4 \right. \\ & + 12bcq^3 + 6bdq^2 + 360\frac{\Delta^4}{\epsilon}(28a^2q^2 + 14abq + 2ac + b^2) \\ & + 12\frac{\Delta^2}{\epsilon}(68a^2q^4 + 68abq^3 + 28acq^2 + 8adq + 15b^2q^2 + 10bcq \\ & \left. + 2bd + c^2) + 4c^2q^2 + 4cdq + d^2 \right)\end{aligned}$$

$$\text{Bias}(g_{\text{naive}}) = 2\frac{\Delta^2}{\epsilon} \left(12a\frac{\Delta^2}{\epsilon} + 6aq^2 + 3bq + c \right)$$

Unbiased

$$g_{\text{unb}}(X) = X^4a + X^3b + X^2c + Xd - 2\frac{\Delta^2}{\epsilon}(6X^2a + 3Xb + c) + e$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^4 + bq^3 + cq^2 + dq + e$$

$$\begin{aligned}\text{Var}(g_{\text{umb}}(X)) = 2\frac{\Delta^2}{\epsilon} & \left(13248a^2\frac{\Delta^6}{\epsilon} + 16a^2q^6 + 24abq^5 + 16acq^4 + 8adq^3 + 9b^2q^4 \right. \\ & + 12bcq^3 + 6bdq^2 + 36\frac{\Delta^4}{\epsilon} (184a^2q^2 + 92abq + 12ac + 7b^2) \\ & + 2\frac{\Delta^2}{\epsilon} (276a^2q^4 + 276abq^3 + 108acq^2 + 24adq + 63b^2q^2 \\ & \left. + 42bcq + 6bd + 5c^2) + 4c^2q^2 + 4cdq + d^2 \right)\end{aligned}$$

$$\begin{aligned}\text{MSE}(g_{\text{umb}}) = 2\frac{\Delta^2}{\epsilon} & \left(13248a^2\frac{\Delta^6}{\epsilon} + 16a^2q^6 + 24abq^5 + 16acq^4 + 8adq^3 + 9b^2q^4 \right. \\ & + 12bcq^3 + 6bdq^2 + 36\frac{\Delta^4}{\epsilon} (184a^2q^2 + 92abq + 12ac + 7b^2) \\ & + 2\frac{\Delta^2}{\epsilon} (276a^2q^4 + 276abq^3 + 108acq^2 + 24adq + 63b^2q^2 + 42bcq \\ & \left. + 6bd + 5c^2) + 4c^2q^2 + 4cdq + d^2 \right)\end{aligned}$$

Comparison

$$\text{mean gap} = -2\frac{\Delta^2}{\epsilon} \left(12a\frac{\Delta^2}{\epsilon} + 6aq^2 + 3bq + c \right)$$

$$\begin{aligned}\text{variance gap} = -24\frac{\Delta^4}{\epsilon} & \left(552a^2\frac{\Delta^4}{\epsilon} + 16a^2q^4 + 16abq^3 + 8acq^2 + 4adq + 3b^2q^2 \right. \\ & \left. + 2bcq + bd + \frac{\Delta^2}{\epsilon} (264a^2q^2 + 132abq + 20ac + 9b^2) \right)\end{aligned}$$

$$\begin{aligned}\text{MSE gap} = -4\frac{\Delta^4}{\epsilon} & \left(3456a^2\frac{\Delta^4}{\epsilon} + 132a^2q^4 + 132abq^3 + 60acq^2 + 24adq + 27b^2q^2 \right. \\ & \left. + 18bcq + 6bd + 18\frac{\Delta^2}{\epsilon} (96a^2q^2 + 48abq + 8ac + 3b^2) + c^2 \right)\end{aligned}$$

$$\begin{aligned}\text{Bias}(g_{\text{naive}})^2 = 4\frac{\Delta^4}{\epsilon} & \left(144a^2\frac{\Delta^4}{\epsilon} + 36a^2q^4 + 36abq^3 + 24a\frac{\Delta^2}{\epsilon} (6aq^2 + 3bq + c) \right. \\ & \left. + 12acq^2 + 9b^2q^2 + 6bcq + c^2 \right)\end{aligned}$$